

**THE LESSONS FROM U.S. ENVIRONMENTAL POLICY AND DISPUTE RESOLUTION
- ONONDAGA LAKE REMEDIATION CASE -**

By

Sang-Yuk Kim

RESEARCH PAPER

Submitted to
KDI School of Public Policy and Management
in partial fulfillment of the requirements
for the degree of

MASTER OF PUBLIC POLICY

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ABSTRACT

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Onondaga Lake is one of the most polluted water bodies in the United States. Its present condition is the result of more than a century of municipal sewage and industrial pollution. Fortunately, it is on the way to recovery under the intergovernmental cooperation and the citizen's involvement. By researching this case, we can find that U.S federal environmental policy interventions to the states and the municipalities through enforcing the Clean Water Act and the Superfund Program as well as facilitating the Alternative Dispute Resolution Mechanisms in the environmental issues have been effective. On top of that, we may understand the importance of the technical and financial assistances from the upper level of governments for alleviating the side-effects of unfunded mandates.

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Dedicated to Soo-Mi Choi

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- TABLE OF CONTENTS -

1. Introduction.....	9
<i>Background</i>	<i>9</i>
<i>Purpose</i>	<i>9</i>
<i>Compositions</i>	<i>10</i>
<i>Method</i>	<i>11</i>
2. Factual Aspects of the Onondaga Lake	11
<i>Geographical Facts</i>	<i>11</i>
<i>Historical Facts</i>	<i>12</i>
<i>Bio-Chemical Facts</i>	<i>13</i>
3. Remedial Endeavors.....	13
<i>A. Background</i>	<i>13</i>
<i>B. Federal Intervention.....</i>	<i>14</i>
<i>Clean Water Act of 1972 and Civil Litigation.....</i>	<i>14</i>
<i>Superfund Program and Brownfield Initiative</i>	<i>15</i>
<i>C. Responsibility of New York State</i>	<i>16</i>
<i>Clean Water/Clean Air Bond Act</i>	<i>16</i>
<i>Superfund Program Implementation</i>	<i>17</i>
<i>D. Roles of Onondaga County as Miniature of Water Pollution Control Agency.....</i>	<i>18</i>
<i>Poor Sewage Treatment</i>	<i>18</i>
<i>Development of a CWA Civil Litigation.....</i>	<i>19</i>
<i>ACJ Requirements</i>	<i>20</i>
<i>Status of ACJ Compliance</i>	<i>21</i>
<i>E. Efforts of Syracuse City.....</i>	<i>22</i>
<i>F. Challenge of Environmental Organization</i>	<i>23</i>
<i>G. Intergovernmental Cooperation</i>	<i>24</i>

<i>Onondaga Lake Management Conference</i>	24
<i>Onondaga Lake Partnership</i>	25
<i>U.S. Army Corps of Engineers</i>	25
4. <i>Evaluation</i>	26
<i>A. Environmental Federalism and Civil Participation</i>	27
<i>Introduction</i>	27
<i>Environmental Federalism</i>	28
<i>Civil Participation</i>	29
<i>Scorecard of EPA-ASLF Intervention</i>	30
<i>Conclusion</i>	33
<i>B. Environmental Dispute Resolution</i>	33
<i>Introduction</i>	33
<i>EDR/ADR System in U.S.</i>	34
<i>EDR/ADR in Onondaga Lake Case</i>	35
5. <i>Midland Dispute to be Solved</i>	36
<i>Introduction</i>	36
<i>What is the Midland Plant Project?</i>	37
<i>Who are Involved? What are the Claims?</i>	37
<i>What Have They Done Until Now?</i>	39
<i>What are the Problems in the Past Endeavors of the Parties?</i>	45
<i>How Can They Resolve the Conflict?</i>	46
<i>Conclusion</i>	48
6. <i>Lessons for Korea</i>	49
<i>Importance of Precautionary Principle and Sustainable Development</i>	50
<i>Pre-Conditions of ADR/EDR Mechanism</i>	51
<i>Technical and Financial Assistance with Municipal Mandates</i>	51
<Exhibit 1> <i>Onondaga Lake Watershed</i>	53
<Exhibit 2> <i>Onondaga Lake Superfund Site</i>	54
<Bibliography>.....	55

THE LESSONS FROM U.S ENVIRONMENTAL POLICY AND DISPUTE RESOLUTION - ONONDAGA LAKE REMEDIATION CASE -

1. Introduction

Background

The Onondaga Lake, located in Onondaga County of New York State, is one of the most polluted water bodies in the United States. Onondaga Lake's present condition is the result of more than a century of municipal sewage and industrial pollution. Fortunately, it is on the way to recovery under the intergovernmental cooperation and citizen's involvement.

The New York State Department of Environment Conservation (NYSDEC) and Onondaga County are doing central role in Onondaga Lake water quality improvement and lakefront remediation. It is, however, impossible to explain in limited perspective of a government or an organization in order to answer the questions of what programs are undergoing, what entities are involved for Remediation of Onondaga Lake, what they have done, what are the problems to be solved.

Most interesting and dynamic part of Onondaga Lake case is, so called, the Midland dispute, a site-specific local environmental conflict. Through this case, we can see past, present, and future issues surrounding not just Onondaga Lake, but also Syracuse and Onondaga County.

Purpose

Most of all, I need to understand and evaluate more practically how the major U.S environmental programs apply in actual case, how the different level of governments interact,

how the governments is checked by the court and civil organization under the 'dreadful' civil litigation clause of CWA, how the environmental conflicts are interacting each other, how useful the ADR are, and how the local governments manage the communities facing environmental disputes.

The other purpose of this research is to get some lessons which are applicable to the environmental policy and management of Korean government. Though Korea has complicated environmental legal system and the Ministry of Environment, environmental protection system in Korea still has a long way to go. Most of public projects for environment remediation and development hardly proceed without strong opposition from the communities, the media, and the environmental organizations.

Compositions

The second chapter is the factual aspects of the Onondaga Lake. The third is the remedial endeavors employed by each governmental entity. This chapter deals with not only current environmental policies but also its applications to Onondaga Lake case.

In the fourth chapter, as a center of this research paper, I do thematic analysis on overall environmental management in terms of 'Environmental Federalism and Civil Participation' and 'Environmental Dispute Resolution'. In the analysis of 'Environmental Dispute Resolution' surrounding Onondaga Lake remediation process, I do a case study on 'the Midland dispute' in separate chapter.

The last chapter, as conclusion of this paper, is about the lessons from environmental management system of the United States and Onondaga Lake case to Korea.

Method

The data were gathered from the publications and the website information of the each government and the related organizations. The news archives of Post-standard¹ were searched and reviewed to follow the Midland dispute. Related federal court documents were reviewed. Relevant academic writings were cited for thematic analysis. The observation and visitation on the lake, lakefront, industrial zone, Metropolitan Syracuse Sewage Treatment Facility and the Midland dispute site were implemented.

2. Factual Aspects of the Onondaga Lake

Geographical Facts

Onondaga Lake is located along the northern end of the City of Syracuse (26 square miles, 146,435 residents, 2000), in Onondaga County (784 square miles, 458,336 residents, 2000), New York.

The Lake is 4.6 miles long and 1 mile wide. Its average depth is 35 feet, and its maximum depth is 63 feet. It has a drainage basin of 248 square miles covering all of the City of Syracuse and many of its suburb towns and villages east and west of the lake.

The major natural tributaries to the lake are Nine Mile Creek and Onondaga Creek. Together they account for 70% of the total amount of water that flows into the lake each year. The Metropolitan Syracuse Sewage Treatment Plant contributes 20% of the lake's annual inflow. Other tributaries are Ley Creek, Bloody Brook, and Harbor Brook. Minor tributaries include Sawmill Creek, Tributary 5A, and the East Flume. Water leaves the lake through a

¹ Central New York based local news paper. <http://www.syracuse.com/poststandard>

single outlet at the lake's north end. This outlet flows to the Seneca River and ultimately reaches Lake Ontario. <Exhibit 1> shows simplified Onondaga lake watershed and the geography surrounding the lake.

Historical Facts

Since explorers had discovered salt springs on the shores of the Lake in 1654, salt production was major industry in early history of the Syracuse city. Late 19th century, Onondaga Lake was a popular resort area. The west shore of the lake had many hotels, parks and bathing beaches. Swimming, boating and fishing were common activities during the summer season.

However, since the Solvay Process Company (later to become Allied-Signal and currently Honeywell International, Inc.) began production of soda ash in 1884, the water quality was aggravated. The company began production of organic chemicals in 1918, chlorine production by the mercury cell process in 1946.

Due to aggravated water quality, swimming was banned in 1940, fishing in 1970. The company closed chlorinated benzene plant and chlor-alkali plant in 1977 and soda ash manufacturing operations in 1986. Catch-and-release fishing was allowed since 1986. Now, Superfund program is being applied on this area.

Syracuse city built sewers and banned backyard privies in 1896. Sewage flew directly into Onondaga Creek and Harbor Brook. In 1960, Onondaga County completed the Metropolitan Syracuse Wastewater Treatment Plant (METRO). Now, the County is cleaning up the Lake reducing the Combined Sewer Overflows (CSOs) and improving the METRO.

Bio-Chemical Facts

Major pollutants of concern can be divided into two main areas. One is those related to wastewater collection and treatment. Another is those related to past industrial operations and uses.

Major pollution problems related to wastewater collection and treatment include high levels of phosphorus in the lake water, high levels of ammonia and nitrite, bacterial contamination of the water after storm events.

Major pollution problems related to past industrial operations and uses include contamination of the water by mercury and other toxic substances, high levels of salinity in the lake water. Other pollution problems of concern include excessive sedimentation, primarily from the Tully Valley mud-boils, and non-point source pollution generated in the lake's watershed.

3. Remedial Endeavors

A. Background

Historically, industrial processing plants and municipal wastewater treatment plants routinely discharged their wastes into the Lake. The necessary measures, however, for cleaning up the Lake were not efficiently initiated by the State and the County.

This situation changed by three circumstantial changes. Firstly, Atlantic States Legal Foundation (ASLF) filed a complaint in 1988 against the Onondaga County alleging violations of Clean Water Act (CWA). And the parties involved this case successfully reached

a consented agreement in form of 'Amended Consent Judgment (ACJ)' in 1997. ACJ mandated the State and County to take key measures to improve water quality of the lake.

Secondly, Environment Protection Agency (EPA) enlisted the Lake as a Superfund site on 'the National Priority List' in 1994. By this measure, the State could take remedial actions to remove hazardous wastes on the lake and lakefront.

Lastly, the economic decline became conspicuous since General Electric's radio and television design and production center and other manufacturing companies were closed or downsized. The County and the city of Syracuse urged to redevelop the lakefront area from abandoned industrial land to recreational area or commercial zone.

B. Federal Intervention

Since 1970, U.S. federal government intervened in the environmental issues of the States and Municipalities by introducing and reinforcing environmental regulations. Clean Water Act and Superfund Act were those. Though the United States Army Corps of Engineers is federal agency, the role will be explained in Intergovernmental Cooperation part.

Clean Water Act of 1972 and Civil Litigation²

EPA revised Clean Water Act, originally enacted in 1948, in 1972 giving its current shape. By CWA, EPA utilizes both water quality standards and technology-based effluent limitations to protect water quality applied to industrial and municipal sources through numerical effluent limitations in discharge permits.

² Copeland, Claudia. 1999. Clean Water Act: A Summary of the Law, CRS Report

CWA requires each state to establish water quality standards for all bodies of water in the state. Control of toxic pollutant discharges has been a key focus of water quality programs. In addition, states are required to implement control strategies for waters expected to remain polluted by toxic chemicals even after industrial dischargers have installed the best available cleanup technologies required under the law.

EPA has oversight of state enforcement and retains the right to bring a direct action where it believes that a state has failed to take timely and appropriate action or where a state or local agency requests EPA involvement.

Furthermore, the federal government acts to enforce against criminal violations of the federal law. In addition, by section 505, individuals may bring a citizen suit in U.S. district court against persons who violate a prescribed effluent standard or limitation. Individuals also may bring citizen suits against the Administrator of EPA or equivalent state official for failure to carry out a non-discretionary duty under the Act.

Thus, as I mentioned above, a local environmental organization sued the County for violation of CWA which accelerated the Lake clean up projects.

Superfund Program³ and Brownfield Initiative

In 1976, Responding to public concern over “midnight dumping” of toxic wastes, Congress established authority for controls over hazardous waste from generation to disposal under the Resource Conservation and Recovery Act (RCRA).

³ USEPA, Office of Solid Waste and Emergency Response. 2000. Superfund: 20 years of protecting human health and environment. <http://www.epa.gov/superfund>

In 1980, Congress passed the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) to address the dangers of abandoned or uncontrolled hazardous waste dumps by developing a nationwide program for: emergency response; information gathering and analysis; liability for responsible parties; and site cleanup. CERCLA also creates a Trust Fund (or 'Superfund') to finance emergency responses and cleanups.

In 1993, The Brownfield Initiative was launched. EPA defines Brownfield as "abandoned, idled, or under-used industrial and commercial facilities where expansion or redevelopment is complicated by real or perceived environmental contamination." By differentiating and identifying Brownfield sites from Superfund sites which implies huge legal liability, the governments and private developers can implement the projects without concerning uncertain costs.

EPA added Onondaga Lake to the Federal Superfund National Priority List in 1995. Onondaga Lake Superfund Site⁴ includes the Lake itself, seven major and other minor tributaries, and upland sources of contamination to the Lake (sub-sites). <Exhibit 2> shows the details of the site. Moreover, as request of Syracuse city, EPA gave a subsidy for establishing 'the Syracuse Brownfield Redevelopment Pilot Program' in 2000.

C. State Responsibility

Clean Water/Clean Air Bond Act⁵

While the CWA addresses federal enforcement, the majority of actions taken to enforce the law are undertaken by states, both because states issue the majority of permits to dischargers

⁴ EPA Regional Office 2. 2002. Superfund Site Description-Onondaga Lake New York

⁵ NYSDEC. 1998. List of Water Quality Improvement Projects to be Funded by the Bond Act.

and because the federal government lacks the resources for day-to-day monitoring and law enforcement. Like most other federal environmental laws, CWA enforcement is shared by EPA and states with states having primary responsibility.

In addition to CWA enforcement, NYSDEC operates state fund under the Clean Water/Clean Air Bond Act. The Bond Act was legislated in November 1996 providing \$1.75 billion for priority environmental projects to ensure the further protection of the State's air, water and natural resources. It authorizes investments in five major program categories: Clean Water projects; Safe Drinking Water projects; Clean Air projects; Environmental Restoration or Brownfields projects; and Solid Waste projects.

The Metro Digital Control System and Aeration System Upgrade Projects of Onondaga County were funded by this state fund in Fiscal Year of 1997~1998. The Bond Act also provided \$1.7 million for brownfield clean up project at Midtown Plaza, a 2.4-acre site at Syracuse.

Superfund Program Implementation⁶

As like CWA, Superfund program implementation is shared by EPA and the State. The State, however, does more active role in Superfund program than in CWA enforcement.

Allied Corporation closed its Syracuse facilities in 1986. On June 27 of 1989, New York State filed a lawsuit against Allied-Signal Inc., alleging past and continuing contamination of Onondaga Lake. The State's claims for remediation and cleanup costs were based on the common law of public nuisance and the New York State Environmental Conservation Law,

⁶ Onondaga Lake Cleanup Corporation. 2001. The Status of Onondaga Lake. Page 22~23

but center on the CERCLA. In January 1992, the State and Allied-Signal, Inc. signed an interim Consent Decree detailing the elements of a Remedial Investigation and Feasibility Study (RI/FS) of the lake to be conducted by Allied-Signal. The RI/FS is required to assess waste-bed related impacts to the lake, in addition to mercury and toxic organic contamination of the lake's ecosystem associated with the company's operations.

Allied-Signal submitted draft reports for most elements of the RI/FS to the State, and the State is reviewing this material. The State is also planning to conduct a Natural Resources Damage Assessment (NRDA) of the lake. When the RI/FS and NRDA are complete, the DEC Commissioner will select the appropriate remedies that need to be implemented in Records of Decision.

The State has entered into a number of consent orders with other Potentially Responsible Parties (PRPs) who may have played a role in the contamination of the Lake.

D. Roles of Onondaga County as Miniature of Water Pollution Control Agency

Poor Sewage Treatment⁷

In 1960, Onondaga County completed Metropolitan Syracuse Wastewater Treatment Plant (METRO). The METRO, however, is currently the lake's main phosphorus and ammonia contributor. Treated domestic and industrial wastes discharged from a pipe at the METRO directly into Onondaga Lake represent a point source input. Between 1990 and 1998, the METRO contributed 65% of the total phosphorus loading to Onondaga Lake. The remaining 35% of phosphorus entered the lake from its natural tributaries. Some of the phosphorus

⁷ Onondaga Lake Partnership. 2001. The Status of Onondaga Lake. Pages 7~10. www.onlakepartners.org

enters the tributaries from non point sources such as runoff carried from the surrounding land.

Other sources of phosphorus to the tributaries are from Combined Sewer Overflows (CSOs) which discharge untreated sewage and storm water to Onondaga Creek, Ley Creek and Harbor Brook during intense rainfall. The sewers in older cities like Syracuse were built to combine storm water and sewage. All wastes that reach the METRO plant are disinfected with sodium hypochlorite which kills bacteria and viruses. However, during heavy storms, excess flow from the sewer system never reaches the METRO plant. Instead, untreated diluted sewage and floating debris are discharged into the tributaries of Onondaga Lake through relief structures called CSOs.

CSOs are also the main source of bacteria to Onondaga Lake. They cause the lake to violate bacteria standards after heavy rainfall. Public health standards for levels of indicator bacteria are routinely violated throughout the southern half of Onondaga Lake following storm events. After a storm, the water in Onondaga Lake remains contaminated with bacteria from the CSOs for about 3 days. During average summer months there are 18 storms that can lead to about 54 days of violations. These violations prevent the lake from being used for swimming.

Development of a CWA Civil Litigation

In 1988, Atlantic States Legal Foundation (ASLF) filed a complaint against the Onondaga County alleging violations of 33. U.S.C. section 301, sub-chapter 1311(a) and 1311(b)(1)(b) in the discharge of sewage to Onondaga Lake from the County's METRO and CSOs and the State intervened as plaintiffs alleging that the defendants had violated the New York State Environmental Conservation Law, section 17-0509 and 17-1511 by failing to provide the level

of treatment required under the law with respect to the discharge of sewage into Onondaga Lake and its tributaries.

After 10 years of endeavors for consensus building, Judge McAvoy, U.S District Court for the Northern District of New York, authorized an agreement⁸ among ASLF, the State and the County.

During the litigation, the County had to perform a series of engineering and scientific studies to evaluate the need for upgrading the METRO and for providing treatment of the CSOs that occur in the Metro service area. Based on the results of those studies and in consultation with the NYSDEC and the USEPA, the County developed a plan for upgrading the Metro plant and addressing the CSOs.

The County submitted the proposed Municipal Compliance Plan (MCP) to the State and ASLF on January 11 of 1996. Following submission of the MCP, there were numerous discussions and negotiations with regard to the proposed MCP. The result was the execution of the Amended Consent Judgment (ACJ) which was signed in January 1998 by all the parties—NYSDEC, the State Attorney General, ASLF, and the County.

ACJ Requirements

ACJ became central part of Onondaga lake improvement masterplan and future legal responsibility, not just policy objectives, of the County.

The improvements undertaken as a part of the ACJ are expected to cost \$380 million (in 1999 dollars) and will be undertaken over 15 years. In the event that future compliance

⁸ ASLF vs. Onondaga County, 88-CV-OO66, Northern District Court of New York. <http://www.onlakepartners.org>

determinations require the construction of additional filtering for phosphorous or a pipe to the Seneca River commencing in the year 2010, the cost of the project would be expected to increase by \$65 million.

Construction costs is expected (not guaranteed) to be supported by direct aid from state and federal governments, low interest subsidized loans from the New York State Environmental Facilities Corporation, and user charges assessed to owners of property located in the Consolidated Sanitary District. The County estimates receipt of direct state and federal aid totaling approximately \$260 million over the 15-year construction schedule. Sources of these funds include \$75 million in direct aid from State Environmental Bond Act funds and \$36 million in federal aid already appropriated for projects included in the plan.

ACJ, however, provides that the state will “endeavor”, consistent with its legal authority, and, where necessary, subject to appropriation, and consistent with its responsibility to promote the health and welfare of its citizens, to obtain state financial assistance to help the County meet its obligation under the ACJ. And the County’s obligations are not conditioned on the availability or receipt of financial assistance from the State or any other entity.

Status of ACJ Compliance⁹

By the end of 2002, the County had completed 22 projects required under the ACJ, about two-thirds of the mandated projects. METRO treatment facilities, \$125 million project, expected to be completed in 2004, will provide advanced treatment at Metro and will further reduce levels of ammonia and phosphorus in the plant's effluent. When the new facilities are operating, Onondaga Lake's water quality should benefit significantly. Concentrations of

⁹ Onondaga County Department of Water Protection, Oct 2003, <http://www.lake.onondaga.ny.us>

ammonia and phosphorus should be noticeably lower resulting in improved fish spawning and survival of juvenile fish. The dissolved oxygen in the lake should be higher, and the water should exhibit better clarity. The endeavors for abating combined sewer overflow also is seeing success by completing most of regional storage and treatment facilities, sewer separation, floatables control facilities, and storage & transport capacity.

The County, however, faced strong opposition of the residents concerning that a regional treatment facility near the Midland avenue in Syracuse would injure their quality of life. This issue will be reviewed in detail in later part.

E. Efforts of Syracuse City¹⁰

The efforts of Syracuse city for environmental friendly urban management are aiming at economic renaissance. As regards traditional industry, the County and City are trying to develop Electrics Park and Hancock Airpark which are located at miles away from the Lake¹¹. These municipalities are planning to redevelop lakefront and old industry zone on Lakefront toward recreational and commercial zone.

Syracuse city initiated 'The Syracuse Brownfields Redevelopment Pilot Program' in 2000. Its purpose is to identify and assess brownfields in the City of Syracuse. The goal is to identify and prioritize five brownfield sites based on perceived threat to the community, development potential and cost efficiency.

The More ambitious plan is 'The Lakefront Development Plan'. 'The Lakefront Development Corporation (LDC)', created by the City in 1996, gets the initiative for

¹⁰ Syracuse city, Oct 2003, <http://www.syracuse.ny.us>

¹¹ Metropolitan Development Association of Central New York, Oct 2003, <http://www.mda-cny.com>

facilitating the redevelopment of the 800 acre area in Syracuse known as the Syracuse Lakefront. The LDC is also working with the City of Syracuse and landowners in the Lakefront Area to create major zoning changes based on the Lakefront Area Master Plan and the concepts of New Urbanism, Sustainable Development and Smart Growth.

The County and the City is now marketing the Lakefront Area as a tourism destination, community centerpiece, urban neighborhood and cutting edge office district. The biggest project is, so called, 'the DestiNY USA'¹². This, 2-billion-mammoth-scale, environmentally friendly tourist development project, was initiated by a private company, but with expectation of the economic incentives from the governments. The City and the County plan to give property tax incentive. The State plans to give benefits of 'the Empire Zones Program'. The project also expects the benefits of 'the Federal Empowerment Zone Program'. Other plans involve 'the Onondaga Creekwalks', 'the Inner Harbor Redevelopment', and downtown business promotion near the Onondaga Creek.

F. Challenge of Environmental Organization

Atlantic States Legal Foundation (ASLF)¹³ filed civil litigation against Onondaga County and successfully got the comprehensive commitment which became a cornerstone for the Onondaga Lake Remediation. ASLF, located in Syracuse, was established in 1982 to provide affordable legal, technical and organizational assistance to individuals, community groups, and other Non-Governmental Organizations (NGOs), as a way to effectively re-mediate threats to the natural environment.

¹² DestiNY USA, Oct 2003, <http://www.destinyusa.com>

¹³ Atlantic States Legal Foundation, Oct 2003, <http://www.aslf.org>

Throughout the 1980s and early 1990s, ASLF was the principal NGO utilizing the citizen suit provision of the Clean Water Act as a means of forcing polluters to cease the desecration of America's waterways.

ASLF is still monitoring the implementation of 'ACJ' and participating in policy making process as a non-voting member of 'the Onondaga Lake Partnership'.

G. Intergovernmental Cooperation

Due to the complexity of the problems of Onondaga Lake, intergovernmental cooperation was necessary. The Onondaga Lake Management Conference and the Onondaga Lake Partnership, tuned by U.S. Army Corps of Engineers as a federal technical assistance, were those examples.

However, it is still not easy to coordinate the partnership due to the fact that the remedial solutions are influenced by a variety of technical, legal, social and political factors. The Midland dispute is one of those.

The Onondaga Lake Management Conference¹⁴

Through a host of federal environmental regulations and enforcement actions, as well as the closing, in 1986, of the Allied-Signal chemical manufacturing complex which was a major contributor of industrial pollutants, the levels of pollution of the Lake have been reduced.

However, the complexity of the Onondaga Lake system and the continued impact of numerous pollutants required a comprehensive and coordinated approach to the goal of

¹⁴ OLMC. 1993. Onondaga Lake: A Plan for Action. <http://www.onlakepartners.org>

reclamation. In response, the U.S Congress created the Onondaga Lake Management Conference (OLMC) in 1990 (Public Law 101-596, section 401) and charged it with responsibility for 'the development of a comprehensive restoration, conservation and management plan' and 'the coordination of the implementation of the plan'.

The OLMC was made up of six voting members: the Assistant Secretary of the Army Corps of Engineers for Civil Works, the Administrator of EPA, the Governor, the State Attorney General, the County Executive and Mayor of Syracuse. It drafted "A Plan for Action" which became the basis of "the Onondaga Lake Management Plan".

The Onondaga Lake Partnership¹⁵

The OLMC grew into 'the Onondaga Lake Partnership (OLP)' with the same 6 voting members but with more non-voting members and extended responsibility. The U.S Congress passed 'the Water Resource Development Act of 1999', which also authorized \$10 million in federal funds toward the Lake clean up effort. The OLP was officially established in 2000. The Partnership is now coordinating development and implementation of improvement projects in accordance with not only the Onondaga Lake Management Plan but also the Amended Consent Judgment to restore, conserve and manage the lake.

The U.S. Army Corps of Engineers¹⁶

The U. S. Army Corps of Engineers (USACE), Buffalo District office, leads the Partnership on behalf of the Secretary of the Army. In 1992, the USACE completed 'the Onondaga Lake Water Quality Technical Report' outlining possible lake remediation alternatives. In fact, the

¹⁵ OLP. Oct 2003. <http://www.onlakepartners.org>

¹⁶ USACE. Oct 2003. <http://www.usace.army.mil>

USACE is an indispensable entity in water resource management in United States. As the 'Federal Engineer', USACE is the largest engineering organization in the world, consisting of 53 regional offices and four engineering research laboratories staffed mostly with civilian employees totaling more than 37,000 professionals in various technical centers, divisions and districts across the nation.

USACE supports civil works projects involving environmental restoration, navigation, flood control, water supply, water quality and emergency responses to natural disasters. USACE works closely with communities, state and Federal during the development of projects to restore and protect water and land resources. USACE is providing engineering assistance to EPA for toxic waste removal under the Superfund program as well.

4. Evaluation

As reviewed above, all the levels of the governments, federal court and a civil organization have affected on the Onondaga Lake remediation. To evaluate the environmental management system surrounding Onondaga Lake Remediation, I would like to raise two questions. The first is that "has the federal intervention and civil litigation really brought good performance for the environment and the quality of life of the residents?" This is so called 'the Environmental Federalism'. The second is that "have the stakeholders solved the environmental disputes in an efficient manner?"

As a matter of fact, though there is financial and technical assistance from the federal government, most of financial and administrative burden to implement the policy is on the State and the County. And 'the Alternative Dispute Resolution (ADR) does not always work.

A. Environmental Federalism and Civil Participation

Introduction

The quality of life is ultimate goal of environmental policy. But no one cannot be sure environmental quality will guarantees quality of life itself since there are so many factors affecting on quality of life. Actually, we need well-designed, safe and quick roads, secured jobs to be happy as well as clean water and air. Those needs, however, are differ from country to country, from state to state, from municipality to municipality in the question of which combination will maximize the happiness of the society. In this point, all the environmental regulation enforced by upper level governments does not mean optimum solution to the community.

In addition, civil participation does not render high quality of policy automatically. Civil participation, especially in environmental management by organized manner, provides the governments with various up-to-date idea, technical research, rational checking and mediation between the government and issue-specific interest group. But occasionally, environmental organizations hinder the governments in deciding optimal policy in whole picture by putting environmental concern always in the first priority. This problem can be worsened in a situation that the civil group is given the legal right of filing an action in the court and asking specific governmental action entailing huge budget expenditure.

The concerns mentioned above can be paraphrased by the questions of which level of government does or should have the final responsibility and accountability for the quality of life and how we can balance and minimize the negative impact of federal intervention and civil participation after scoring gains and losses qualitatively from them. This part introduces

some debates surrounding environmental federalism and civil participation and tries to score environmental intervention of federal-civil coalition through considering Onondaga Lake improvement endeavors given by all the level of governments and one representative environmental organization, ASLF.

Environmental Federalism

Federalism is the 'relationship between a central government and lower units of that government', where 'the central government and the lower units exercise independent authority within their own jurisdictions'. Therefore the term itself means intergovernmental model of policy implementation.

But the term 'regulatory federalism' has been used pejoratively to describe a perceived trend in national legislation, especially since the 1960s, toward imposition of regulations on state and local government, which are mandatory, far-reaching in impact, and involve substantial un-reimbursed costs for affected jurisdictions¹⁷.

Historically, environmental policymakers justified a strong federal role on the basis of several considerations: (1) the implications of interstate or trans-boundary pollution which individual states could not effectively control; (2) fear that states competing for economic development will engage in a "race to the bottom," if left to their own devices; (3) concern that a hodgepodge of separate state regulation will impose financial costs on out-of-state producers operating in national markets; and (4) belief that administrative efficiency and economies of scale can be achieved by developing complex technical standards at the national level.

¹⁷ M. S. Hamilton, 1990, Regulatory Federalism, Natural Resources and Environmental Management, pp.26

Environmental policymaking typically takes place in the context of a system with several levels of government. And this raises the important issue of the appropriate role of the various governments in the setting of environmental standards, the design of regulatory measures to attain the standards, and the monitoring and enforcement of these measures. Oates says that when environmental quality is local public good or there is spillover effect, unified standards do not maximize environmental quality¹⁸.

In similar perspectives, Blodgett¹⁹ says critics of environmental regulatory federalism assert these new mandates are too often expensive, inflexible, inefficient, inconsistent, intrusive on state prerogatives, ineffective in achieving social goals, and unaccountable. And they questions which level of government might best determine the extent of environmental protection and which is most likely to select the least costly methods of control. He says the answers will dictate moving the locus of decision making to states and localities.

Civil Participation

Desalio & Langton says citizen participation challenges to technocracy in order to ensure democratic policy making²⁰. Actually, federal regulatory agencies like EPA with little political supporters cannot help collaborating civil environmental organization facing reluctant sub-national governments and resisting regulated private entities not only because civil participation is needed to boost democracy but also press the implementation of the regulations.

¹⁸ Oates, Wallace E. 2001. A Reconsideration of Environmental Federalism

¹⁹ Blodgett, John E. 2000, Environmental Protection: New Approaches, CRS Report

²⁰ DeSario, Jack & Langton, Stuart. 1987. Citizen Participation in Public Decision Making, pp 5~17

Moreover, EPA and pro-environmental congressmen give a legal standing as a plaintiff to environmental organizations in some statutes such as the Clean Water Act. Consequently, they use civil action as a most effective way to attain their goals. Through litigation, they can check the mistakes of governments, ask specific projects, monitor implementation status and get the legal cost back from tax. But that is only procedural democracy. That does not guarantee substantial democracy including equality.

Scorecard of EPA-ASLF Intervention

Then, specifically, what have the Onondaga County, its residents, and neighbors gained from federal regulations and the litigation with ASLF?

First of all, apparently, that is improvement in environmental quality. There are many indicators that water quality of Onondaga Lake was improved. According to the County, there are conspicuous progresses in bacteria levels, near-shore clarity, phosphorus loading, algae and water clarity, ammonia and nitrite concentrations, dissolved oxygen and biodiversity²¹.

Secondly, by the result of litigation, civil participation was formalized in the whole implementation processes. Actually, ASLF has a major oversight role over 15-year lifetime of the agreement. It includes reviewing all technical aspects of the agreement's implementation, sampling and analyzing water quality, overseeing environmental expenditure, and facilitating public involvement²².

²¹ Onondaga County, Department of Water Environment Protection. 2003. Onondaga Lake Ambient Monitoring Program Executive Summary, 2002 Annual Report. Pages 10~12

²² ASLF Newsletter, Fall 1997.

Thirdly, in the process of long endeavors to improve Onondaga Lake, a model Partnership was established. Through OLP, the community could get technical and financial assistance.

Then, what have the Onondaga County, its residents, and neighbors lost from federal regulations and the litigation with ASLF?

Firstly, the County had to take much administrative burden to come up with and analyze alternatives for settling down the litigation. The conditions of the Judgment on Consent required the County to perform a series of engineering and scientific studies to evaluate the need for upgrading the Metropolitan Sewage Treatment Plant and for providing treatment of the combined sewer overflows.

But it seems that the first place where the County must concentrate its administrative power is economic development in the situation that there is general decline of regional economic activity, the loss of manufacturing jobs, and decline of local sales tax revenue for local government²³.

Secondly, there is too much financial burden for implementation of 'Unfunded Mandates' without any enough official guarantee of financial assistance from federal and state governments. Total \$ 380 million in one purpose project through 15 year is huge amount. It is about 45% the amount of the County's one-year budget expenditure²⁴. The Policy Research Center of Syracuse University also predicted this problem. It said, "The County will face a growing debt burden if it chooses to continue using general obligation debt to finance

²³ Central New York Regional Planning and Development Board, 2002, CNY Comprehensive Economic Development Strategy

²⁴ Onondaga County, 2003, 2003 Onondaga County Annual Budget

wastewater infrastructure and if it fails to obtain state aid to cover an appreciable portion of remediation costs”²⁵.

Thirdly, fixing when/where/what should be done are inevitably followed by shadow on the decision since technology does change and other policy constraints and context are uncertain in the future. The proposed projects for ACJ were stipulated after environmental impact assessment on land, water, air, plant, animal, agricultural resources, aesthetic resources, historic & archaeological resources, open space & recreation, transportation, energy, noise and odor, public health, and growth & character of community or neighborhood.

However, in terms of impact on open space & recreation, location of Metro Treatment Facility can be questioned after DestiNY project was emerged on the lakefront near the facility. And in the stage of implementation, the technology used and location of sewer abatement facilities are challenged by the residents since impact on neighborhood was underestimated.

Lastly, federal intervention and participation of environmental organization hinder local democracy by limiting flexibility in budget process and unorganized participation through local legislature. Now, we can see the conflict between the County mandated to implement ACJ and Syracuse City responsible to the citizens surrounding the Midland avenue sewage facility since 1999. Affected people are living on Syracuse's South Side in a low-income, mostly minority residential neighborhood along the Midland Avenue. This conflict resulted in two litigations, legitimacy of financial assistance of EPA to the project²⁶ and condemnation of

²⁵ Duncombe, William et al. 1997. The Economic and Fiscal Impact of Lake Remediation on Onondaga County. Center for Policy Research, Syracuse University, Syracuse, New York

²⁶ ASLF and Syracuse United Neighbors vs. USEPA, December 2000, Northern District Court of NY

city-owned property²⁷, finalized with the victory of the County. But this conflict was evitable under normal process of local democracy.

Conclusion

In conclusion, we cannot finalize the debate of Environmental Federalism and Citizen Participation. But we see there are gains and losses in Onondaga Lake case. For the municipalities, it seems that the future tasks are maximizing the gains and minimizing losses. To do so, the most important thing is to keep flexibility in budgeting, project planning, and using technology through incessant communication among every level of governments and every form of civil groups.

In Onondaga Lake case, the losses as a result of CWA and Un-funded Mandate by EPA & ASLF coalition, fortunately, does not seem crucial. Because the County's long-term development strategy is, to some extent, related to sustainable development of lakefront. Other local governments can learn some lesson from this point. Simply put, local government can raise its resident's quality of life by making the best of improved environment whether that is achieved by their own will or mandated by upper level government.

B. Environmental Dispute Resolution

Introduction

One of the most difficult environment management issues is solving site-specific types of environmental disputes in small communities.

²⁷ ASLF vs. Onondaga County, 88-CV-OO66, November 2002, Northern District Court of New York

In the stage of the environmental policy-making, the federal and State governments cannot help considering uniform way of regulation so that they can enforce the policy without discrimination. The courts have tendency to make order to be over-specific, inflexible and irreversible way.

But the municipalities should consider every single voice, property right, legal aspect and every single technical aspect as well as their own long-term development strategies when implementing the policy and order so that they implement the policy or order in more practical, efficient and equitable way. So, nowadays, the proper management of environment disputes resolution (EDR) is one of the most difficult tasks for local governments.

EDR/ADR System in U.S²⁸

It is considered that the traditional EDR approaches, such as avoiding without talks, contending without alternatives, quick-fixing without long term solution and litigation with lot of time, cost and loss of control, are not 'Resolution' in narrow meaning. Nowadays, EDR is considered as the same meaning of Alternative Dispute Resolution (ADR).

ADR consists of a set of techniques, process, and roles that enable parties to a dispute to reach agreement with the help of a neutral third party known as mediator²⁹. The third party's help can be classified into various processes. Arbitration, Convening, Facilitation, Fact Finding, Mediation, Mini-trial, Neutral Evaluation, Partnering and Settlement Judge.

²⁸ EPA. 1999. Status Report on the Use of Alternative Dispute Resolution in EPA Enforcement and Site-Related Actions (Through Fiscal Years 1997 & 1998)

²⁹ Rosemary O'Leary et al. 1999. Managing for the Environment. Page 193. Jossey Bass

Federal laws and policies facilitating ADR cover Administrative Dispute Resolution Act, Alternative Dispute Resolution Act, Civil Justice Act, The Executive Order on Civil Justice Reform, Department of Justice Guidance on the Use of ADR and National Performance Review and Guidance on the Use of ADR in EPA enforcement cases.

EPA utilizes ADR to facilitate the settlement of conflicts related to environmental enforcement actions and site-related disputes. EPA has utilized ADR to facilitate the settlement of civil actions under a broad range of authorities, including the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), Resource Conservation and Recovery Act (RCRA), Clean Water Act (CWA), and Clean Air Act (CAA). Early ADR efforts primarily involved CERCLA cases. But recently the use of ADR in non-CERCLA cases has become equally common. Civil actions have been assisted primarily through convening and mediation neutrals, though other ADR processes have been used. Finally, EPA has included ADR processes in the dispute resolution provisions of administrative and judicial settlement documents.

In addition to the governmental laws and policies, U.S ADR system is backed by extensive ADR infrastructure. The infrastructure consists of a lot of ADR service organizations like American Arbitration Association, public ADR specialists in and out of EPA, ADR centers in Academia, ADR curricula and Mediation License System and so on.

EDR/ADR in Onondaga Lake case

Initially, EDR/ADR system in United States was well functioned in Onondaga Lake case, especially concerned with the dispute of sharing liability on the superfund site and the civil litigation filed by ASLF. However, in the dispute surrounding Midland Avenue Regional

Treatment Facility (RTF) for CSOs, the stakeholders have not reached any constructive agreement. It seems that there is initial difference in the characteristics of the cases, which makes the Midland disputes difficult to be solved even by ADR mechanism.

O'Leary³⁰ indicates the conditions for success of ADR; there are a manageable number of identifiable interests; there are no major issues that involve scientific uncertainty; the core dispute raises no fundamental value and symbolic issues; the parties are able to achieve a level of trust and a sense of shared purposes; there exists a rough parity in relative power among the participants.

Unfortunately, it seems that the Midland dispute does not satisfy any of the conditions. But there seems to be more things to try to solve the problem. I will deal with this issue in detail changing the chapter.

5. Midland Dispute to be Solved

Introduction

This dispute mainly based on the different view on its environmental/economic impact of an environmental processing facility on the neighborhood. We can say this conflict stemmed from 'ACJ' because this facility is one of mandated projects on the basis of 'ACJ'.

Now the conflict between Onondaga County and the regulation coalition of federal government, State government and ASLF shifted to the inter-municipal conflict between the County and the City.

³⁰ Rosemary O'Leary et al. 1999. Managing for the Environment. Pages 212~216. Jossey Bass

Kenneth Klase³¹ says, “Small communities have numerous characteristics such as ‘financial conditions’, ‘limit human and technical resources’ and ‘time availability’ which affect their capability to respond to environmental needs”. And he says, “These characteristics increase the vulnerability of the community to new environmental regulations and have a detrimental effect on local environmental service provision”. It does not seem that the County is able to overcome Krase’s limitation.

What is the Midland Plant Project?

The Onondaga County is trying to construct \$60 million ‘Midland Combined Sewer Regional Treatment Facility (RTF)’ near Oxford Street and Midland Avenue, south side of the Syracuse City from 1999 aiming 2007 as part of the County's combined sewer overflow (CSO) abatement program. The plant is expected to store, separate and disinfect overflowing raw sewage, then release clean water to Onondaga Creek.

Who are Involved? What are the Claims?

The main parties in this conflict are the County charged in construction of the plants and resident’s group involving the Syracuse United Neighbors (SUN), the Onondaga Nation, and the Partnership for Onondaga Creek, claiming the plants will have negative environmental/economic impact on the low-income African American neighbors. But specifically speaking, EPA and the State side on the County by financing the project partly. On the other hand, the City, City Council, Onondaga County Legislator delegated from the district and ASLF side on SUN by filing litigations claiming that EPA cannot grant the

³¹ Klase, Kenneth. 1995. The Problems of Designing Environmental Mediation for Small Communities. J. Walton Blackburn and Willa M. Bruce, Mediating Environmental Conflicts: Theory and Practice.

County before extensive environmental impact survey and the County cannot make a condemnation the City owned lands necessary to the plant construction.

The opposition coalition raises a bunch of claims that: the football field sized facility is too big to be sited in residential area and will have negative environmental impact on the neighbors; the recreational area such as playground and green area will be shifted into the concrete building; the environmental impact will lower the property value near the facility; some of the residents do not want to move from their longtime habitat and friends; there was environmental racism in deciding the site; the part of the project, such as creek walkway, will breach the historical property of the Onondaga Indian Nation³²; the right of access by the affected residents and the City was disregarded; this project cannot guarantee the clean water of Onondaga Lake despite of huge investment of the tax; the County should look for and evaluate alternative sites or alternatives technology involving, but not limited to, underground construction; the project may be inconsistent with the County's long-term development plan.

On the other hand, the County responds that: the site is proved to be most appropriate one through sufficient engineering review; the plant is for the emergency situation, not year long operation; the resident's concern on environmental emission involving 'chlorine' are not found in other similar facilities operated in the County; open space will be provided by putting the recreational and aesthetic consideration in master-plan; most of sewage treatment facilities are connected each other meaning that independent modification of site and technology employed cannot be taken; this project is inevitable since it is the court order,

³² ASLF vs. USEPA. March 2000. Declaration of Chief Irving Powless, JR of Onondaga Nation. Northern District Court of New York

otherwise, the tax will go for the penalty; the County has not much time to comply to the court order and not enough money available; the neighbor's sacrifice will be compensated by the County's impact easement plan; this project and other Onondaga Lake water quality improvement plan will have long-term benefits for the community.

What Have They Done Until Now?

The controversy between parties started in March 1999 still goes on. The substantial progress for splitting the difference of each other's position is not found. But the opposition coalition raised two litigations and has done various informal efforts. The County made a progress in the water quality improvement by successful implementation of other mandated projects and has studied to come up with their best offer for Midland project by modifying construction design and proposing impact easement plan. And the parties have been held meeting and negotiations without solid resolution. Let's keep track of their efforts.

April 1999, Some Syracuse Common Council members said they could not support the Midland project after hearing objections from residents. Council members asked county to search for some other alternatives.

January 2000, Onondaga County received federal permission to start construction of the Midland plant. The U.S. Environmental Protection Agency ruled the project would not have any "significant environmental impacts" other than stopping the flow of about 300 million gallons of raw sewage into Onondaga Creek each year.

January 2000, Onondaga County proposed a \$3 million plan to ease the impact. County offered to relocate 36 families, build housing and develop a walkway along the Creek to make

the plan more acceptable to residents. The plan included nine townhouses, a park with a basketball court, a playground and green space along with a creek-walk.

March 2000, a coalition opposed to the Midland Avenue sewage plant sued the federal government, seeking to halt the project until a search for alternatives could be completed. Lawsuit filed on behalf of 23 South Side residents, business owners, religious institutions and community groups.

July 2001, Second Circuit Court of Appeals in Manhattan affirmed a lower court ruling that the Midland project may proceed and receive federal financial aid. The coalition of South Side groups sought to block Onondaga County from receiving more than \$45 million in aid for the project from the U.S. Environmental Protection Agency.

August 2001, a study commissioned by Syracuse officials failed to find cheaper alternatives to building the Midland plant. Engineers hired by the city said it would cost about \$138 million for an alternative plan to end sewage overflows into Onondaga Creek by separating sanitary sewers from storm sewers.

October 2001, Syracuse Common Council refuses to sell land the county wants for construction of the Midland plant. To the cheers of a standing-room-only crowd, the council casts unanimous "no" votes on seven pieces of legislation the county needs to build the plant.

March 2002, engineering consultants for the County found that building six underground sewage storage tanks in the neighborhood could be an alternative but would cost \$15 to \$30 million more than the original proposal. SUN showed interest but the County worried whether affordable or not.

August 2002, dramatically the parties had tried to reach an agreement by three days negotiation with the mediation of the State. But it ended in vain since the County could not certain on the cost and technical detail of underground storage.

November 2002, Northern District Court of New York ruled that the County has authority to condemn the city-owned properties at issue.

February 2003, Onondaga County Legislature voted to let the County Administration buy or condemn the properties. And the County came up with an alternative that the plant would move 60 feet farther from a row of town houses along Blaine Street. But the City and resident have paid little attention.

March 2003, the first and latest face-to-face talks, between Onondaga County Executive Nicholas Pirro and Syracuse United Neighbors, ended without constructive progress.

Now, it seems that the County takes its own way to meet the deadline by fixing the budget on the project. The residents, however, still on the road raising the pickets.

Then, how can we get the clues to resolving this old but imminent dispute? To answer this question, we need to understand the source and nature of this conflict and what should the BATNA looks like. Next, we need to reflect the past trajectory of the dispute and evaluate the parties' past/current efforts to make it out.

E. What's the source of conflict? What are characteristics of the conflict?

Susan Senecah says, "The environmental conflicts arise from different values, different stakes and different assessments of the environmental policy/management and its risks". And

Guy and Heidi Burgess³³ say, “It become aggravated by confused interests, technical disagreements, misunderstanding, questions of procedural fairness and psychological escalation and polarization”.

As regards the characteristics of environmental conflicts, Susan Senecah³⁴ writes, “Typically, the environmental conflicts are multi-party involved, resource disparate, multi-objective, science and technology based, steeped in laws and regulations, contentious and high profile, consensus building process daunting”.

Similarly, Rosemary O’Leary³⁵ also wrote that “the characteristics of environmental conflicts include: multiple forums for decision making; inter-organizational conflicts; multiple parties; multiple issues; technical complexity and scientific uncertainty; unequal power and resources; and public/political arenas for problem solving”.

It seems that above discussions fit well in this case. It is clear that the various parties in their nature involved in Midland Plant dispute have all different political and economic interests. The County Executive Nicholas Pirro and majority of County Legislators cannot sacrifice majority of taxpayers for the low-income minority in one municipality. On the other hand, The City Mayor and Councilors will feel challenging pressure to their political capacity in their own community. Likewise, overall county taxpayers will not want the County to spend money more than legal obligations. But it is obvious that the residents living near the

³³ Burgess, Guy and Burgess, Heidi. 1994. Beyond limits; Dispute Resolution of Intractable Environmental Conflicts. Walton Blackburn and Willa Bruce. Mediating Environmental Conflicts: Theory and Practice.

³⁴ Senecah, Susan. 2000. Practical Theory for the Environmental conflict Assessment Process: The Trinity of Voice, Legitimacy, and Influence

³⁵ O’Leary, Rosemary. 1995. Environmental Mediation: What Do We Know and How Do We Know It? Walton Blackburn and Willa Bruce, eds. Mediating Environmental Conflicts: Theory and Practice

site think the compensation by the laws does not cover their sacrifice for the public. Moreover nobody, even engineering consultant, know whether the technical alternatives really will work, how much it will cost and who will pay it in the future.

One more aspect we should not overlook, in the discussion about the characteristic of the conflict, is that this is a kind of identity-based conflict necessarily involving interest-based nature between not only ethnics but also communities.

Actually Onondaga Nation claims the County should not touch their historical land; the City claims it deserves more money for safer plant than the County will spend and the organization was excluded in the process of decision-making; most seriously, SUN accuses the County of 'environmental racism' claiming that it will hurt low-income African American neighborhood without any prior involvement in site selection and technology determination.

Bowen and Wells³⁶ say, "There is not enough empirical evidence supporting 'environmental racism'. But important thing is how the minorities feel. In there, identity-based conflict arises overlapped with interest-based conflicts."

Robert Bullard³⁷ defines 'environmental racism' as "any policy, practice, or directive that intentionally or unintentionally differentially impacts or disadvantages individuals, groups, or communities based on race or color; as well as the exclusionary and restrictive practices that limit participation by people of color in decision making boards, commissions, and staffs." More broadening term is 'environmental justice'.

³⁶ Bowen, William & Wells, Michael. The Politics and Reality of Environmental Justice, Public Administration Review, November/December 2002, Vol. 62, No. 6

³⁷ Bullard, Robert. 1993. Anatomy of Environmental Racism and Environmental Justice Movement, 15-40

Ronald Bass³⁸ defines it as the fair treatment and meaningful involvement of all people – regardless of race, color, national origin, or income – in the development, implementation, and enforcement of environmental laws. It can be interpreted to mean that environmental hazards should be distributed equally across society, and no individual, group, or community should bear a disproportionate burden from the adverse effects of such hazards. It is difficult to say there is equally distributed environmental hazard by Midland Plant.

Jay Rothman³⁹ says that “when people’s essential identities, as expressed and maintained by their primary group affiliations, are threatened or frustrated, intransigent conflict almost inevitably follows. However, in such conflicts, conventional methods of conflict management are usually inadequate and may even exacerbate the problem”. He also mentions that “Interest-based disputes are usually concrete and clearly defined and the outcomes each side seeks are bounded by the resources at stake. However many other conflicts are relatively intangible and deeply rooted in the more abstract and interpretive dynamics of history, psychology, culture, values, and beliefs of identity groups”.

Putting together above discussions, we can understand what should the BATNA looks like in Midland conflict. First, the BATNA should be scientifically right and technically practical one. Second, it should be economically just and affordable one. Third, the parties should be bound to the rule of laws. Fourth, the BATNA should be negotiated and agreed on the political and social consensus. The consensus does not mean an agreement among disputants. More broadly, the BATNA should be consistent to the long-term benefit of the community.

³⁸ Bass, Ronald. 1988. Evaluating Environmental Justice under the National Environmental Policy Act. Environmental Impact Assessment Review: 83-92.

³⁹ Rothman, Jay. 1997. Resolving Identity-Based Conflict. San Francisco: Jossey - Bass Publishers

It seems that the County can do the work without consensus. But that will be never constructive and will produce emotional crevice between the ethnics, class, and communities.

What are the Problems in the Past Endeavors of the Parties?

Next, we need to evaluate the parties' past and current efforts to make out the dispute. By doing so, we can get some hints about what each parties should to do more.

There is no doubt that the County has full legal support involving the court order, upper level government's financial support and regulations, the guaranteed condemnation authority to city-owned parcels. And this aspect is the biggest obstacle the opposition coalition confronts. The County also has tried to come up with the best offer as long as technical feasibility is secured. And it seems that the County Legislature is thinking about the balanced distribution of the County budget. The legislature may see that the \$3 million easement plan also economically unjust to the residents of other municipalities.

But the County cannot deny that the City (32%, 147 thousand population of the County) and the affected residents were not fully involved in the project from the start of the plan before 1997 though the area at issue is sociologically very sensitive one. And it is empirically true that the economic value of properties near the environmental facilities is negatively affected.

Meanwhile, the opposition coalition has spent too much time trying to legally block the proceeding of the plant instead of building practically negotiable alternatives in terms of technical aspect as well as financial aspect. The opposition coalition may earn some time until next election by filing litigations, lobbying politicians and tabling negotiations without any

plausible cards. But it seems that the core position of the County will not be changed. And it is possible that majority of the constituency will support the Midland Plant and the Onondaga Clean-Up Project necessary to attain the County and City's long-term economic development goals through 'DestiNY' project and Lakefront development plan. Actually, this dispute is not 'Development vs. Environment' but 'Majority vs. Minority'.

In summary, it seems that the parties' efforts made little difference from Carpenter & Kennedy's 'conventional approaches to managing conflict'. They⁴⁰ classified conventional approaches into 4 categories, that is: avoiding the issue; leaping into battle – holding public meetings, appealing to the news media, mounting letter-writing campaigns, lobbying politicians, staging public demonstrations, eventually initiating litigation; finding quick fix; the Solomon trap – spending time without progress for implementation in setting tentative solution, attacking it and defending it.

How Can They Resolve the Conflict?

Then, how can it be resolved? And why should it be resolved constructively? Legally speaking, the County can do the work without permission of the City and the residents. But empirically speaking, it is not easy and not desirable.

In the small community, intergovernmental cooperation and private involvement are keys to the successful policy implementation. Once the relationship of trust and mutual recognition would be broken, it takes long time to recover.

⁴⁰ Carpenter, S. L. and Kennedy, W.J.D. 1988. *Understanding Public Disputes: The Spiral of Unmanaged Conflict*. Jossey-Bass Publishers, San Francisco.

Practically, the City has many administrative authorities, which can obstruct the Plant construction. And the residents also have many legal rights, which can be used to obstruct the Plant construction and operation. Moreover, it is often the case that a psychological frustration and anger, caused by over-escalated conflict, result serious incidents or casualties, which can lead the community to disorder. Therefore, it should be resolved constructively, if possible, before the commencement of the construction.

Carpenter suggests 'the alternative approaches' to the conflict manager. He suggest that, "to achieve the goal of consensus decision making and cooperation in implementing solution, conflict manager as a facilitator oversee assessing the problem and bring parties together to exchange information and seek solutions."

Jay Rothman⁴¹, similarly, introduces 'ARIA Framework' as a resolution process to especially identity-based conflict like the case at issue. He says "this framework is designed to address the problem of inadequate and inaccurate information exchange between disputants by contributing to the evolution of openness and an atmosphere of joint inquiry and learning".

And he says "the groundwork must be laid by enabling disputants to give voice to their deepest concerns and essential motivation, and to recognize those of their adversaries." He says, in this framework, the parties should stop 'Antagonism', increase 'Resonance' of expression, develop 'Invention' of mutually acceptable option and take 'Action' what should be done. Then, who should be the conflict manager or who should provide a good office for ARIA Framework?

⁴¹ Rothman, Jay. 1997. Resolving Identity-Based Conflict. San Francisco: Jossey - Bass Publishers

NYSDEC tried to mediate the County and the City at the three days negotiation in August 2002 since DEC understands the legal and technical context of the Midland Plant and Onondaga Clean-up Project as well as has the authority to extend the deadline of compliance to ACJ.

ASLF also can be act as a co-mediator or fact-finder with comparable knowledge to DEC. It is strange that ASLF sides on the resident group since ASLF requested and agreed with the County to implement the project. In fact, it is obvious that ASLF has some responsibility to intervene the disputes as a neutral third party.

But ASLF did little role. It is not too late to help the direct parties in honestly and inexpensively assessing whether or not confrontations with their associated power contests offer more desirable alternatives to negotiated agreements.

It is desirable that, before opening final co-mediation before the county contract out for the construction, they reach a consensus on the answers to: whether the County can afford the underground storage or not; what's economic impact on the neighborhood; how the environmental impact looks like at other plant; how much money the County can spend on additional impact easement plan; when the project should start; what can be resident's role through the engineering, construction and operation stages.

Conclusion

Midland conflict, in conjunction with previous ASLF litigation case, shows us how uncontrollable obligations, burdened to the poor resourced local governments by upper level governments and the court, develop and aggravate the inter-communal conflict. The County

and resident groups have been inevitably restrained in their negotiation choices by un-requested or pre-requested project's inflexible schedule, engineering and site. The affected groups have resisted to the government with a lot of reason.

In every like-situation, it is not justifiable that the governments criticize such a response as a NIMBY (Not In My Back Yard) attitude. Instead, the disputants had better recognize each other's interest and identity with sympathy and full provision of information. ARIA Frame provides a good tool in this sense. Likewise, the parties should understand the nature of issued conflict and least requirement of the BATNA before making the matter worse, so that they can figure out practical options and maintain the neighborhood relationship.

The communal environmental conflict, like any other types of conflicts, is not pleasant experience for the government officials and its citizens. It is true that the local government officials spend a significant time in dealing environmental dispute without feeling of productivity. The residents also sacrifice much of time and cost to get best offer when they feel they are not recognized and not reasonably compensated. However, the odds are that Midland conflict is not only a demanding challenge but also a good opportunity for the municipalities as well as the residents in a sense that they may learn how to resolve the conflict constructively and who can provide a good office in specific case for the future.

6. Lessons for Korea

Onondaga Lake remediation case involves various environmental issues. Those issues cover Superfund and Brownfield program, water quality management, environmental federalism, civil litigation of CWA, environmental justice and intergovernmental partnership.

The extensiveness of issues on Onondaga Lake and the dynamic interaction among the stakeholders give us various lessons. Most of lessons were mentioned earlier in thematic analysis part. The following lessons are especially for the environment policy and management of Korean government and municipalities.

Importance of Precautionary Principle and Sustainable Development

Even though Onondaga Lake remediation can be found to be a successful case, the citizen will not give a high score to the governments. It costs too much for the governments and the community to complete planned remediation projects in Superfund, Brownfield, and Sewage treatment improvement projects.

Historically, early industrial site and old urban area were operated and occupied without serious consideration of environmental security and remediation cost at the end of its life. The results were thousands of polluted Superfund/Brownfield site to be cleaned, financial/administrative burden of the governments, and endless litigation to share the liability with the companies which is potentially guilty.

Fortunately, Korea adopted 'the Precautionary Principle' in 'the National Environmental Policy Act' at early stage of industrialization. Precautionary measures, however, have been implemented poorly. The government will have to employ all precautionary measures in monitoring the industrial activities, operating municipal waste facilities and sewage treatment, and guiding citizen's daily life to be environment-friendly. In the long perspective, Korea will have to pursue sustainable development in designing 'Eco-Industrial Parks' and 'Eco-City'. Otherwise, the environmental risk and economic burden is likely to be carried by the public at the present and in the future.

Pre-Conditions of ADR/EDR Mechanism

Recently, ADR mechanism and civil litigation system in environmental law is taking high attention in Korea. The preconditions, however, have not been satisfied. And there is little research concerning the side effects of civil litigation and pre-conditions to facilitate the ADR.

ADR in U.S is supported and modulated by various federal statutes, executive orders, and EPA guides. There are also a lot of channels of public/private ADR services and training/certification programs such as the American Arbitration Association, the State Dispute Resolution Associations and the Community Dispute Resolution Centers in collaboration with the government agencies, the courts, NGOs and the academia. So, to introduce public litigation system in environmental laws and facilitate ADR, Korea should prepare refined ADR mechanism which is practically applicable to real world.

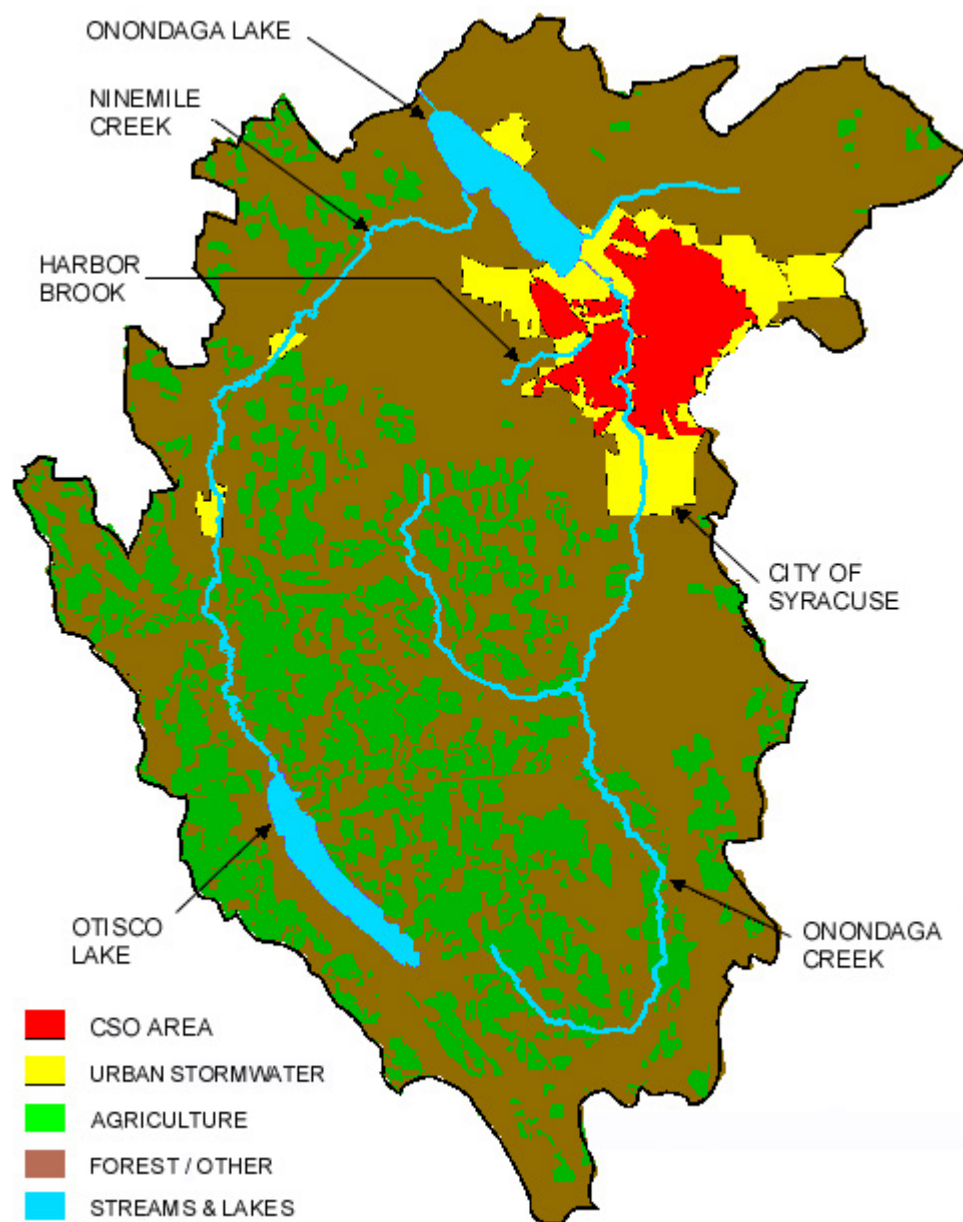
Civil litigation is likely to affect on resource allocation of municipalities. To minimize the negative side effects of civil litigation, the municipalities should be able to come up with flexible and feasible alternatives to be agreed. This will needs the local governments of high technical capacity and financial management skills. The central government should be able to guide the municipalities.

Technical and Financial Assistance with Municipal Mandates

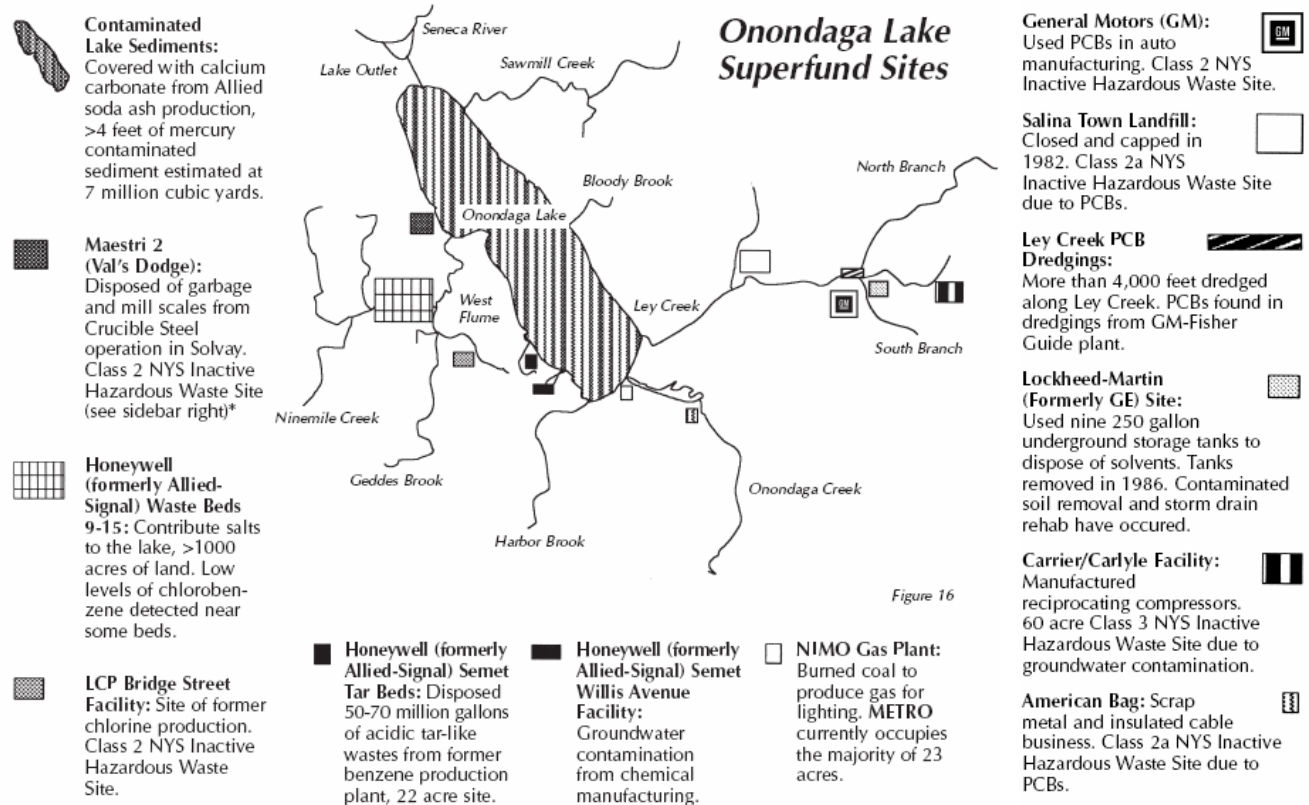
It is important for the environmental protection agencies to give financial and technical assistance to the municipalities as well as enforce the regulations and require mandates to the local governments. Financial assistance, however, is not that easy because overall national budget is limited. The environment agenda will have to be higher national/municipal priority.

An interesting thing I could find in Onondaga case about federal technical assistance is the role of the Army Corps of Engineers. In Korea, there is no public engineering entity. Most of engineering services for the governments are provided by private companies through contracting out. The contracting out, however, could not provide timely and consistent services. Korea has broad and rich military resources in every regional place. The government should make best of this resources. If the people can consider environment as one of security issues, the technical potential of the military and its regional network can be used in environmental protection in peaceful circumstances.

<Exhibit 1> Onondaga Lake Watershed



<Exhibit 2> Onondaga Lake Superfund Site



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